

SRM IST

School of Computing

Department of Networking and Communications

21CSC315J - FOG COMPUTING

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Course Content

Unit 1 - Introduction

Fog Computing Basics, Fog and Edge Computing Completing the Cloud,

Hierarchy of Fog and Edge Computing, Business models, Opportunities and

Challenges, Fog Computing Communication Technologies.

CO1: Contrast Fog Computing and Edge Computing

Fog Computing Basics

Fog computing, also known as fogging, is a decentralized computing paradigm that extends cloud computing by bringing data storage, processing, and applications closer to the devices and users in the network. It plays a critical role in managing the massive amount of data generated by the Internet of Things (IoT), edge devices, and real-time applications.

Fog Computing Basics

Key Concepts of Fog Computing

Proximity to End Users

- Fog computing operates closer to the source of data, such as IoT devices, sensors, or edge devices.
- It reduces latency by processing data locally rather than sending it to a distant central cloud.

Fog Computing Basics

Decentralized Architecture

Unlike traditional cloud computing, fog computing uses a distributed approach where data processing happens across multiple nodes, including gateways, routers, and devices.

Security and Privacy

By keeping sensitive data closer to the source, fog computing can enhance data security and privacy compared to transferring everything to the cloud.

Fog Computing Basics

Real-Time Data Processing

Fog computing is ideal for time-sensitive applications like autonomous vehicles, smart grids, industrial automation, and healthcare systems, as it can process data in real-time or near real-time.

Interoperability

It supports diverse devices and protocols, enabling seamless integration between IoT devices and cloud services.

Fog Computing Basics

Key Components of Fog Computing

Fog Nodes

Devices such as routers, switches, gateways, or dedicated servers that perform data processing, storage, and networking tasks

IoT Devices

Sensors, actuators, and other devices generate data that fog nodes process.

Fog Computing Basics

Cloud Layer

The central cloud serves as a backup and performs complex analytics, storage, and processing tasks when necessary.

Network Connectivity

Communication between fog nodes, IoT devices, and the cloud happens through wired or wireless networks.

Fog Computing Basics

Advantages of Fog Computing

Reduced Latency

By processing data locally, it minimizes delays in decision-making

Bandwidth Efficiency

Only filtered or necessary data is sent to the cloud, reducing network congestion

Fog Computing Basics

Enhanced Security

Sensitive data can remain local, reducing exposure to potential breaches during transit.

Scalability

The distributed nature of fog computing allows it to scale efficiently with the growing number of IoT devices.

Fog Computing Basics

Energy Efficiency

By processing data locally, energy consumption associated with data transmission to the cloud is reduced.

Fog Computing Basics

Challenges of Fog Computing

Complexity

Managing a decentralized system requires robust coordination and infrastructure.

Security Concerns

While it enhances privacy, securing distributed fog nodes against cyberattacks is challenging.

Fog Computing Basics

Resource Limitations

Fog nodes may have limited processing power and storage compared to cloud data centers.

Standardization

The lack of universal standards can hinder interoperability across different systems and devices.

Fog Computing Basics

Applications of Fog Computing

Smart Cities

Traffic management, smart lighting, and waste management.

Healthcare

Real-time patient monitoring and diagnostics.

Industrial IoT

Predictive maintenance, process optimization, and safety systems.

Fog Computing Basics

Autonomous Vehicles

Vehicle-to-infrastructure (V2I) and vehicle-to-vehicle (V2V)

communication for navigation and safety.

Agriculture

Precision farming and environmental monitoring.

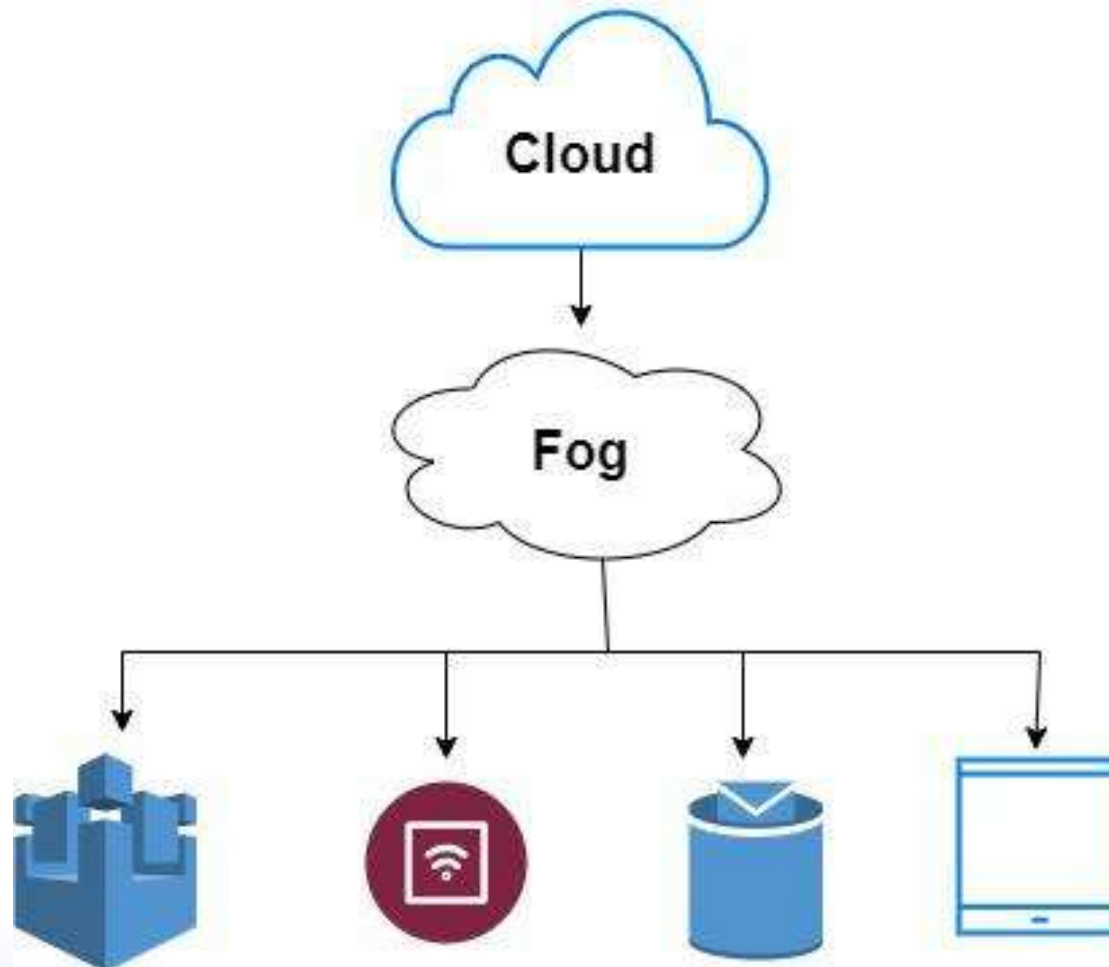
Fog Computing Basics

Fog Computing vs. Edge Computing

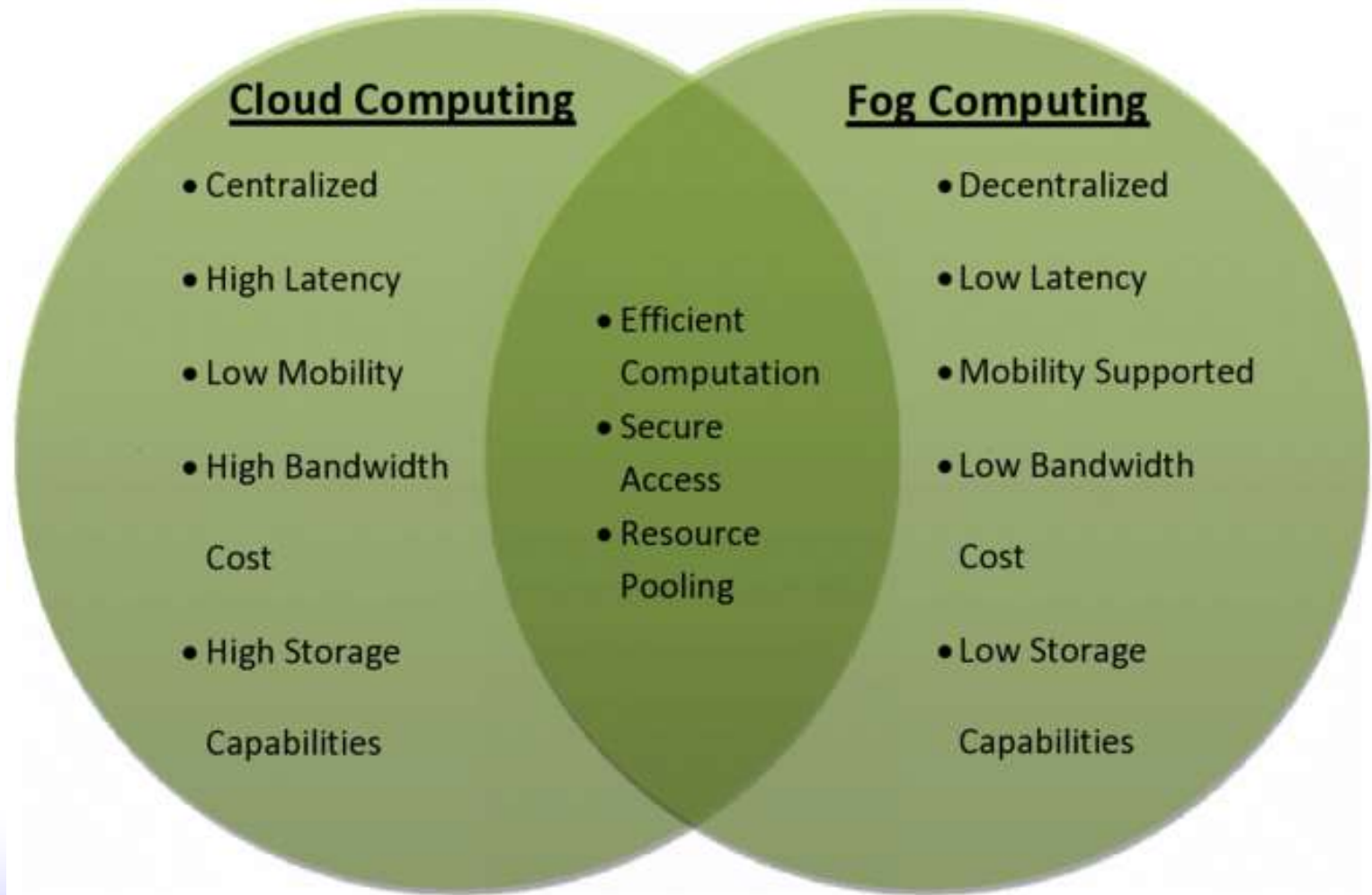
- **Edge Computing** is a subset of fog computing, focusing on processing data directly on IoT devices or close to the source.
- **Fog Computing** includes edge computing but also involves intermediate nodes like gateways and routers, offering more flexibility in data processing and storage.

Fog computing complements cloud computing and edge computing, enabling efficient, secure, and scalable systems for modern connected technologies.

Fog Computing Basics



Fog Computing Basics



Fog and Edge Computing Completing the Cloud

Fog and Edge Computing: Completing the Cloud

Fog and edge computing work together to complement cloud computing, creating a robust ecosystem that addresses the limitations of a purely cloud-based architecture. This combination ensures efficient data processing, enhanced scalability, reduced latency, and better support for IoT and real-time applications.

Fog and Edge Computing Completing the Cloud

Cloud Computing: Strengths and Challenges

Cloud computing provides centralized storage and powerful computational resources, making it an essential component of modern IT infrastructures. However, it has limitations:

Latency: Transmitting data to a distant cloud data center and back can introduce delays, unsuitable for time-sensitive applications.

Cost: Constant data transmission to the cloud can lead to higher costs.

Fog and Edge Computing Completing the Cloud

Bandwidth Bottlenecks: The increasing number of IoT devices generates massive data, overloading network bandwidth.

Data Privacy: Sensitive information is at risk during transmission and storage in centralized locations.

To overcome these challenges, fog and edge computing serve as complementary layers.

Fog and Edge Computing Completing the Cloud

Edge Computing: Bringing Computation to the Source

Definition: Edge computing processes data directly on the device or near the data source, minimizing the need for transmission to distant cloud servers.

Key Characteristics

- Low latency due to on-device or nearby processing.
- Ideal for devices with limited connectivity.
- Supports localized decision-making.

Fog and Edge Computing Completing the Cloud

Examples

- Smart cameras processing video feeds locally for object detection.
- Industrial IoT devices detecting anomalies in machinery in real time.

Fog and Edge Computing Completing the Cloud

Fog Computing: Bridging the Edge and Cloud

Definition: Fog computing serves as an intermediary layer between the edge and the cloud, extending processing and storage capabilities to network nodes like gateways, routers, and servers closer to the data source.

Fog and Edge Computing Completing the Cloud

Key Characteristics

- Distributed architecture, ensuring data processing at intermediate nodes.
- Reduces the data volume sent to the cloud by filtering or pre-processing information.
- Supports complex analytics and resource-intensive tasks that edge devices can't handle.

Fog and Edge Computing Completing the Cloud

Examples

- A gateway in a smart city aggregating data from multiple IoT sensors and transmitting only essential insights to the cloud.
- Fog nodes optimizing traffic signals based on real-time traffic patterns.

Fog and Edge Computing Completing the Cloud

How Fog and Edge Computing Complete the Cloud?

Distributed Processing

- ❖ **Edge Computing** handles immediate, localized tasks (e.g., real-time decision-making).
- ❖ **Fog Computing** performs more complex pre-processing and aggregation tasks at intermediate nodes.
- ❖ **Cloud Computing** handles long-term storage, large-scale analytics, and historical data analysis.

Fog and Edge Computing Completing the Cloud

Improved Latency

- ❖ Edge devices and fog nodes reduce latency by processing data closer to the source, ensuring faster responses.
- ❖ This is critical for applications like autonomous vehicles, remote surgeries, and financial trading systems.

Fog and Edge Computing Completing the Cloud

Optimized Bandwidth Usage

- ❖ Fog computing reduces data volume by pre-processing it before sending relevant insights to the cloud.
- ❖ Edge computing eliminates the need for constant cloud communication for localized tasks.

Fog and Edge Computing Completing the Cloud

Enhanced Scalability

- ❖ The layered approach accommodates the exponential growth of IoT devices and data generation.
- ❖ Cloud resources can focus on high-level analytics, while fog and edge handle local needs.

Fog and Edge Computing Completing the Cloud

Increased Security and Privacy

- Sensitive data can be processed locally on edge devices or fog nodes, reducing exposure during transmission.
- Compliance with data residency and privacy regulations is easier with localized processing.

Fog and Edge Computing Completing the Cloud

Unified Ecosystem: Practical Applications

Smart Cities

- **Edge:** Streetlights automatically adjust brightness based on real-time pedestrian movement.
- **Fog:** A local gateway aggregates data from multiple sensors to optimize city-wide traffic flow.
- **Cloud:** Historical traffic data is analyzed to plan infrastructure improvements.

Fog and Edge Computing Completing the Cloud

Healthcare

- **Edge:** Wearable devices monitor heart rates and issue immediate alerts.
- **Fog:** Hospital servers aggregate patient data from various devices for localized analytics.
- **Cloud:** Centralized systems store and analyze long-term health records for predictive insights.

Fog and Edge Computing Completing the Cloud

Industrial IoT (IIoT)

- **Edge:** Sensors detect machine anomalies and halt operations in real-time.
- **Fog:** Gateways aggregate data for analysis and send actionable insights to the cloud.
- **Cloud:** Historical analysis helps predict maintenance schedules.

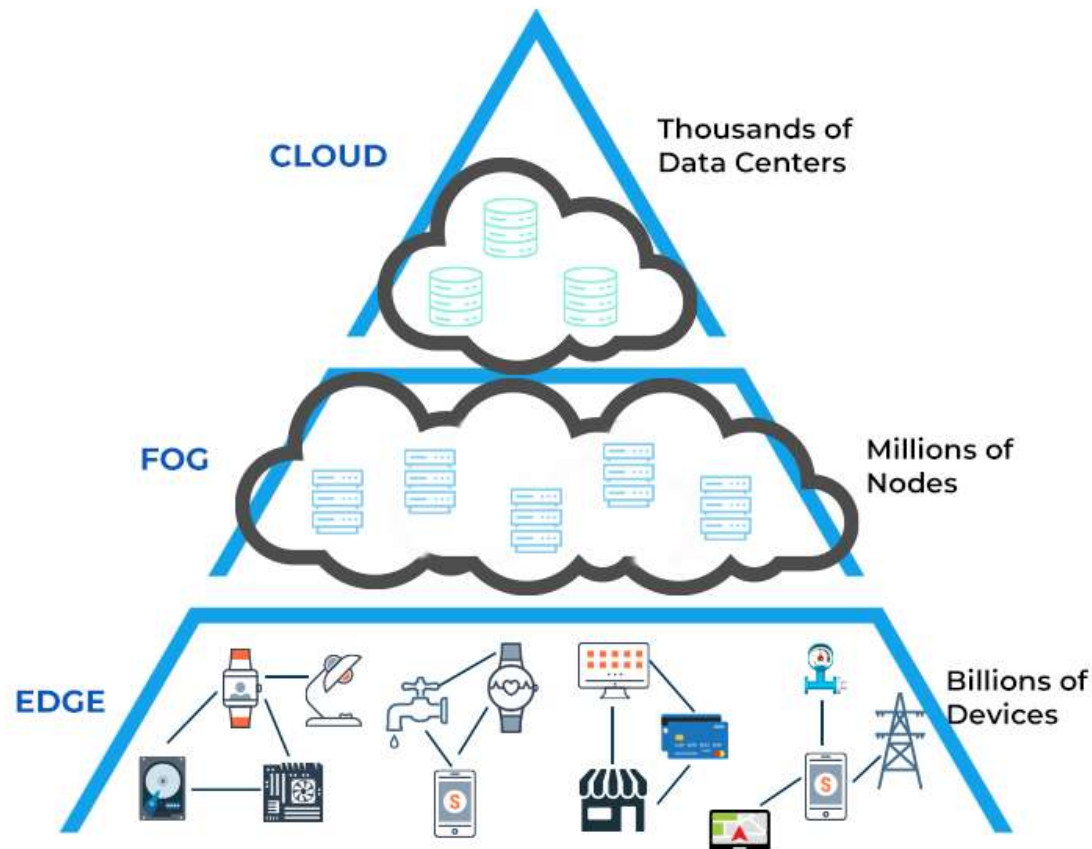
Fog and Edge Computing Completing the Cloud

Conclusion

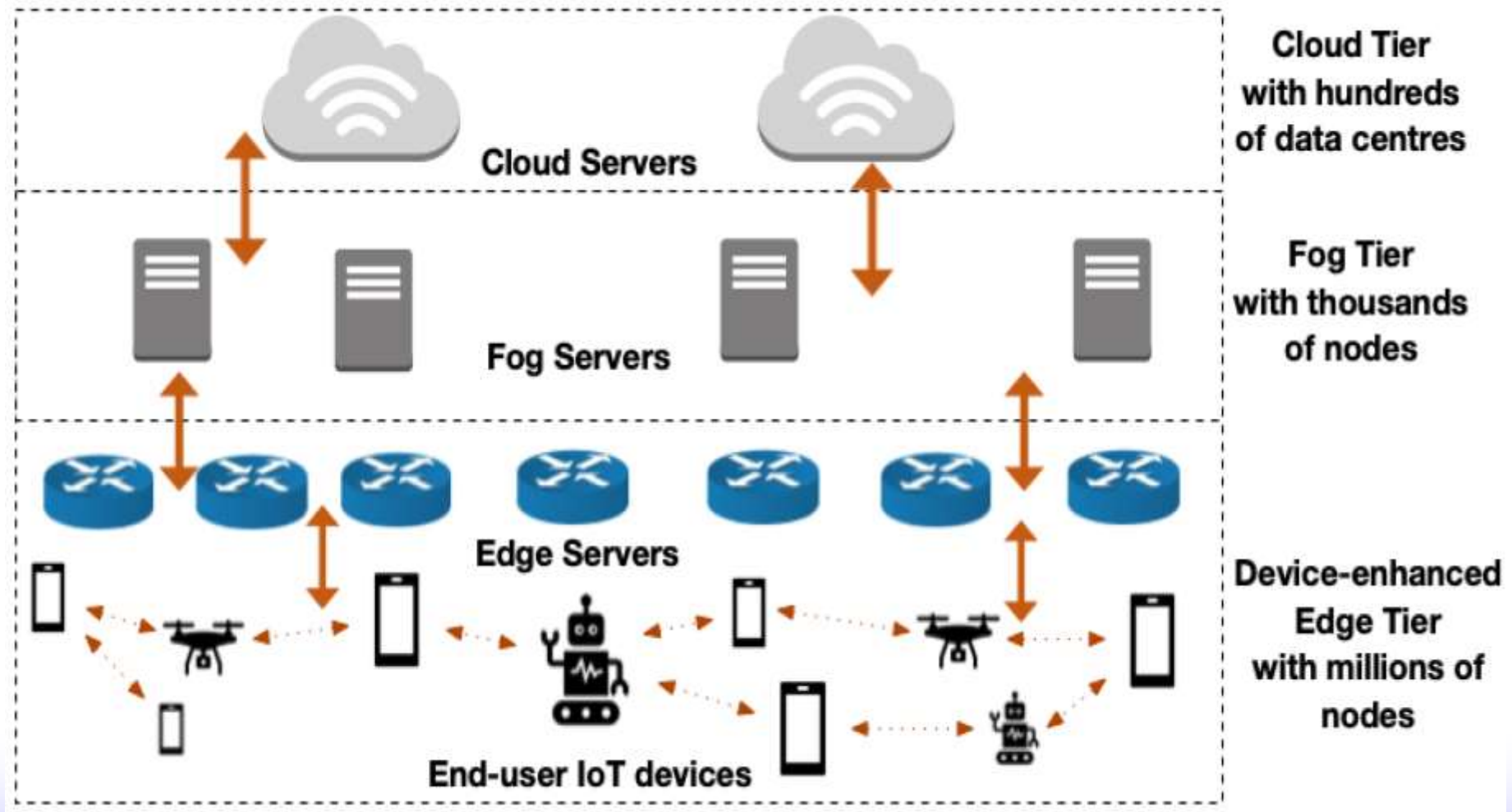
Fog and edge computing, working alongside the cloud, create a distributed, efficient, and secure ecosystem for modern technology. By leveraging the strengths of each layer edge for immediate actions, fog for intermediate processing, and cloud for high-level tasks this combination ensures seamless support for the demands of IoT, real-time applications, and big data analytics. Together, they enable smarter, faster, and more reliable systems, paving the way for the next generation of digital innovation.

Fog and Edge Computing Completing the Cloud

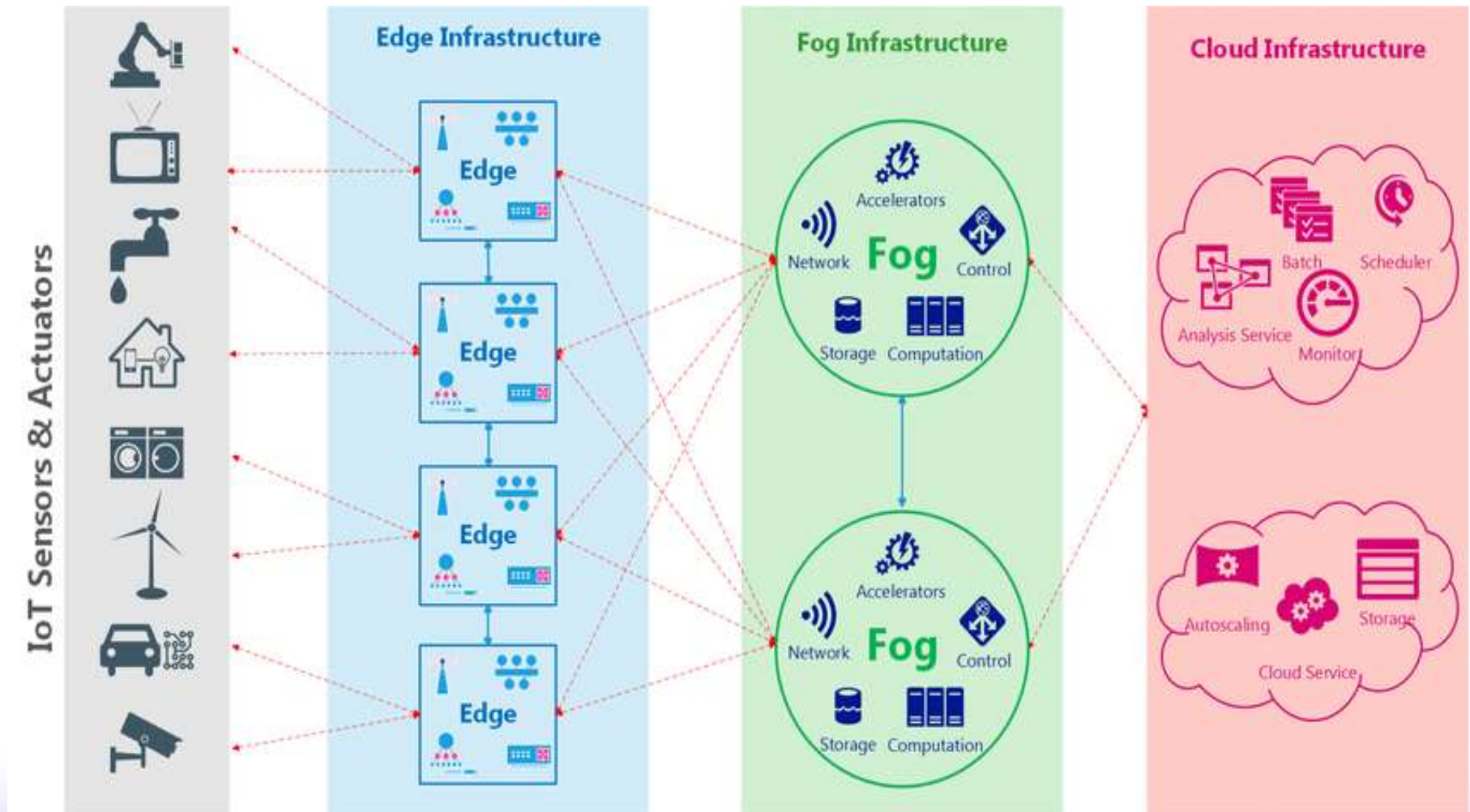
CLOUD COMPUTING VS. FOG COMPUTING VS. EDGE COMPUTING



Fog and Edge Computing Completing the Cloud



Fog and Edge Computing Completing the Cloud



Hierarchy of Fog and Edge Computing

The hierarchy of fog and edge computing represents how data processing, storage, and communication are distributed across various layers in a network. This hierarchy bridges the gap between the cloud and the data sources (IoT devices), enabling efficient, scalable, and real-time data handling.

Concept: How computing resources are arranged from **Cloud** → **Fog** → **Edge** → **Devices** for faster, smarter IoT services.

Why Hierarchy is Needed in IoT

- Traditional **Cloud-centric IoT (CIoT)** faces major issues:
 - **Bandwidth overload** (huge sensor/video streams)
 - **High latency** (real-time apps cannot wait)
 - **Network interruptions** (unstable connectivity)
 - **Resource-constrained devices** (battery + weak processing)
 - **Security challenges** (edge devices vulnerable)
- Hence, computing is pushed closer to data sources using **Fog + Edge Computing**.

What is Fog + Edge Computing (FEC)?

- **Fog and Edge Computing (FEC)** creates a **service continuum** between:
 - **Cloud (centralized)** and
 - **Things / IoT devices (distributed)**
- It enables processing at multiple levels based on need:
 - quick filtering near device
 - analytics near gateway
 - heavy processing in cloud
- This approach improves:
Security, Cognition, Agility, Latency, Efficiency (SCALE)

Hierarchical View of Fog and Edge Computing

- FEC is deployed in **multiple layers**, moving from Cloud outward:
 - **Main FEC layers**
 - **Inner-Edge**
 - **Middle-Edge**
 - **Outer-Edge (Extreme Edge / Mist)**
- This layered structure helps decide *where computation should happen* depending on:
 - latency requirements
 - processing complexity
 - network availability
 - cost and scalability

Overall Architecture

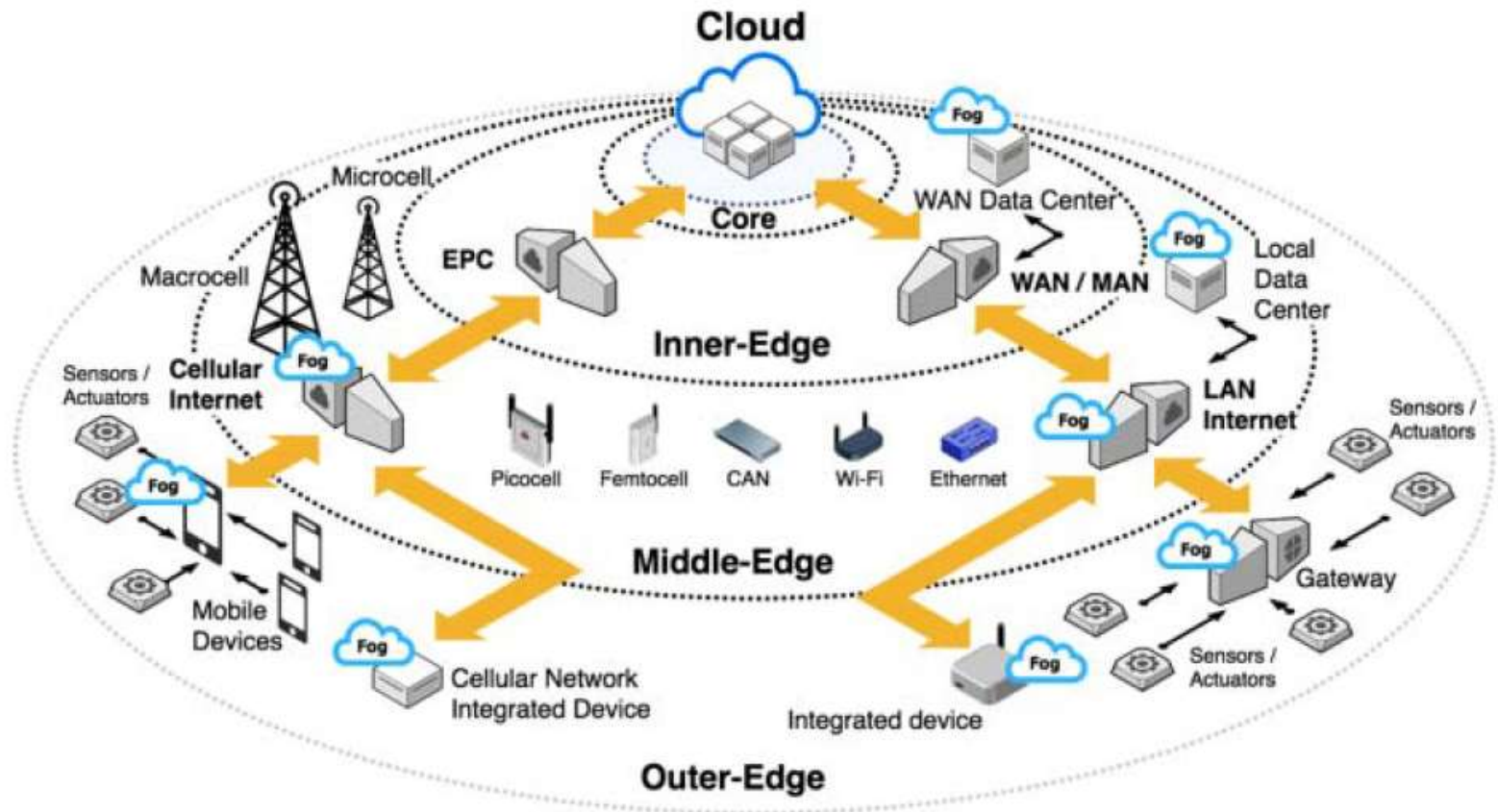


Figure 1.3 Hierarchy of fog and edge computing.

Layer 1 – Inner-Edge (Near-the-edge)

Definition

Inner-edge is the layer closest to the cloud but still **outside the central datacenter.**

Where it exists

- **Enterprise WAN**
- **ISP WAN networks**
- **EPC data centers** (telecom core)
- **MAN (Metropolitan Area Network)**

What it provides

- Geo-distributed **caching**
- Localized **processing**
- Better **QoE (Quality of Experience)**

Example

- Google Edge Network deploying servers inside ISP data centers to speed up services.

Inner-edge can be viewed as “fog at WAN level.”

Why Inner-Edge Matters

- Inner-edge is useful when we need:
 - **Regional low-latency computing**
 - Reliable processing compared to device/gateway
 - Reduced load on central cloud
- **Typical use-cases**
 - streaming acceleration
 - distributed caching
 - regional analytics
 - ISP-based edge cloud services

Layer 2 – Middle-Edge

- **Definition**

- Middle-edge is the *most commonly understood FEC layer*, located between cloud and IoT devices.

- **Major environments**

- **Local Area Networks (LANs)**
- **Cellular Networks**

- Middle-edge supports a wide variety of infrastructure and is highly practical for real deployments.

Middle-Edge (LAN-based Fog/Edge)

- **LAN includes**

- Ethernet LAN
- WLAN (Wi-Fi)
- CAN (Campus Area Network)

- **Typical FEC hosting devices**

- Internet gateway routers
- Industrial routers (example: Cisco fog routers)
- Local server machines in same subnet (cloudlets/local datacenters)

- **Key advantage**

- **1-hop proximity** from IoT devices → very low latency processing.

Middle-Edge (Cellular Network-based Edge)

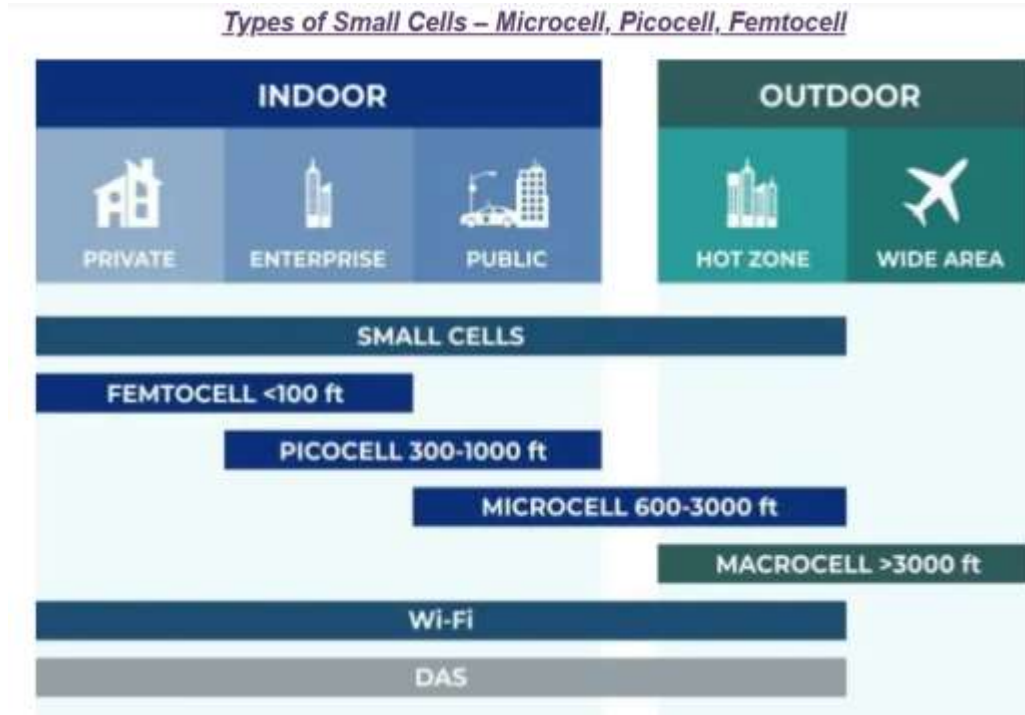
It uses existing telecom infrastructure:

- **Macrocell**
- **Microcell**
- **Picocell**
- **Femtocell**

Why cellular edge is powerful

- Wide-area coverage in cities
- Supports mobile IoT applications like:
 - connected vehicles
 - VR/AR
 - mobile healthcare

MEC (Multi-access Edge Computing) makes this practical.



Layer 3 – Outer-Edge (Extreme Edge / Mist)

■ Definition

- Outer-edge is the **front-end of IoT network**, closest to the real physical world.

Also called:

- extreme-edge
- far-edge
- mist computing

■ It contains 3 main device types:

- Constraint devices
- Integrated devices
- IP gateway devices

Outer-Edge Type 1 – Constraint Devices

■ What they are

- Sensors and actuators
- Microcontroller-based devices

■ Characteristics

- Very limited CPU and memory
- Not suitable for heavy computation
- Can still be updated/reconfigured remotely

■ Why important?

Mist computing focuses on:

- self-management and local decision-making
- strong M2M interaction without full cloud dependence

Outer-Edge Type 2 – Integrated Devices

■ Examples

- Smartphones
- Tablets
- Embedded ARM devices

■ Features

- decent processing and storage
 - built-in sensors (gyroscope, accelerometer)
 - Wi-Fi + Bluetooth support
- These devices can support local processing and act as mini-edge nodes in some use cases.

Outer-Edge Type 3 – IP Gateway Devices

■ What they do

- Act as a bridge between:
- constraint devices (Bluetooth/ZigBee/Z-Wave) and
- IP network / cloud

■ Why gateways are ideal FEC nodes

- Linux based and powerful enough
- supports lightweight virtualization (containers)

■ Examples

- Raspberry Pi
- Hubs
- IoT gateways running Docker/containers

Business Models of Fog Computing

- Fog computing is not only a technical concept.

It also requires **clear business models** to explain:

- how fog services are provided
 - who owns the resources
 - how revenue is generated
- Business models define **value creation and service delivery** in fog computing.

Why Business Models are Important in Fog Computing?

- Fog computing involves:
 - multiple stakeholders
 - distributed infrastructure
 - shared ownership of resources
- Business models help to:
 - define responsibilities
 - support sustainable deployment
 - enable large-scale adoption
 - encourage industry participation

Fog Computing Business Models According to the reference material, **fog computing supports three basic business models:**

- **X as a Service (XaaS)**
 - **Support Service Model**
 - **Application Service Model**
- These models are derived from recent fog and edge computing research.

Business Model 1 – X as a Service (XaaS)

- **X as a Service (XaaS)** means providing fog resources as services.
- Here, **X** can be:
 - Infrastructure
 - Platform
 - Software
 - Networking
 - Storage or Caching
- Similar to cloud service models, but offered at fog/edge level.

XaaS – How it Works

- In this model:
- Fog providers own the fog infrastructure
- Clients pay to use fog resources
- Resources support SCANC mechanisms:
 - Storage
 - Compute
 - Acceleration
 - Networking
 - Control
- Payment is usually **pay-as-you-use**.

XaaS – Key Characteristics

- On-demand resource usage
- Pay-per-use pricing
- Can be offered by:
 - ISPs
 - cloud providers
 - individuals or small businesses
- Supports flexible fog deployment

Business Model 2 – Support Service Model

- In the **Support Service Model**:
- Clients own the fog hardware
- Service providers offer:
 - software installation
 - configuration
 - updates
 - maintenance
 - technical support
- Focus is on **managing and maintaining fog environments**.

Support Service Model – How it Works

In this model:

- Clients purchase fog-enabled devices
- Support providers manage fog software
- Services may be:
 - monthly
 - annual
 - contract-based
- Similar to IT infrastructure support services.

Support Service Model – Key Characteristics

- Hardware owned by clients
- Software and management handled by service providers
- Highly customized solutions
- Suitable for enterprises and industries
- Ensures reliable operation of fog systems.

Business Model 3 – Application Service Model

- In the **Application Service Model**:
- Service providers deliver **complete fog-based applications**
- Clients do not manage fog infrastructure directly
- Focus is on **end-to-end solutions**, not infrastructure.

Application Service Model – How it Works

- Application service providers:
 - deploy fog applications across edge layers
 - process data near data sources
 - provide ultra-low latency services
- Examples:
 - real-time traffic control
 - industrial monitoring
 - digital twin applications

Application Service Model – Key Characteristics

- Complete solution delivery
- Low latency processing
- Fog nodes handle real-time analytics
- Cloud used for long-term processing

Clients pay for **application outcomes**, not infrastructure.

Comparison of Fog Business Models

Model	Ownership	Revenue Source	Focus
XaaS	Provider	Resource usage	Infrastructure & platform
Support Service	Client	Service contracts	Maintenance & management
Application Service	Provider	Application services	End-to-end solutions

Benefits of Fog Business Models

- enable industry participation
- support scalable fog deployment
- reduce dependency on centralized cloud
- improve service quality
- support real-time IoT applications

Simple Example

Smart Traffic System

- XaaS → fog resources provided at traffic junctions
- Support Service → maintenance of fog nodes
- Application Service → real-time traffic control application

Multiple business models can coexist.

Opportunities and Challenges in Fog Computing

Introduction

- Fog computing extends cloud computing to the **edge of the network**.
- While fog offers many **new opportunities**, it also introduces **technical and operational challenges**.
- Understanding both is important for real-world deployment.

Why Study Opportunities and Challenges?

- Fog computing systems are:
- distributed
- heterogeneous
- dynamic
- deployed close to users and devices

So, designers and organizations must:

- understand benefits
- identify limitations
- plan solutions carefully

Major Opportunities in Fog Computing

- Fog computing creates opportunities in:
- performance improvement
- real-time applications
- cost efficiency
- security enhancement
- new business models

Opportunity 1 – Low Latency Processing

- Fog computing processes data **close to data sources**.
- Benefits:
 - faster response time
 - real-time decision making
 - reduced delay compared to cloud

Important for:

- smart traffic systems
- healthcare monitoring
- industrial automation

Opportunity 2 – Reduced Bandwidth Usage

- Fog nodes:
- filter data locally
- process only useful data
- send reduced data to cloud

Benefits:

- lower network congestion
- reduced cloud bandwidth cost
- efficient data handling

Opportunity 3 – Improved Reliability

- Fog computing:
- reduces dependency on central cloud
- supports local processing even during network failure

Useful in:

- remote locations
- unstable network environments
- mission-critical systems

Opportunity 4 – Enhanced Security and Privacy

Fog allows:

- local data processing
- reduced data exposure to cloud
- faster security updates

Benefits:

- improved data privacy
- better protection of sensitive information
- faster threat response

Opportunity 5 – Support for IoT and Smart Systems

Fog computing enables:

- large-scale IoT deployment
- real-time analytics
- smart city and smart industry applications

Fog bridges the gap between **IoT devices and cloud**.

Opportunity 6 – New Business Opportunities

Fog computing supports:

- XaaS models
- application service models
- support service models

Enables participation of:

- telecom providers
- enterprises
- small businesses
- individual service providers

Challenges in Fog Computing

Major Challenges in Fog Computing

- Despite opportunities, fog computing faces challenges in:
- system complexity
- standardization
- management
- security
- deployment and maintenance

Challenge 1 – System Complexity

Fog environments include:

- cloud
- multiple fog layers
- IoT devices

Challenges:

- complex architecture
- difficult coordination
- increased design effort

More layers → more complexity.

Challenge 2 – Heterogeneity

Fog nodes differ in:

- hardware
- operating systems
- network protocols
- processing power

Challenges:

- interoperability issues
- difficult software deployment
- compatibility problems

Challenge 3 – Lack of Standardization

Fog computing:

- is still evolving
- has limited universal standards

Challenges:

- vendor lock-in
- limited portability
- integration difficulty

Standardization is still in progress.

Challenge 4 – Security Management

Fog nodes are:

- geographically distributed
- often deployed in open environments

Security challenges:

- physical tampering
- insecure communication
- distributed attack surface

Securing all fog nodes is difficult.

Challenge 5 – System Management

Fog systems require management across:

- design
- implementation
- runtime adjustment

Challenges include:

- deployment of applications
- monitoring performance
- dynamic reconfiguration

Challenge 6 – Resource Constraints

Some fog and edge devices have:

- limited CPU
- limited memory
- limited storage
- limited power

Efficient resource management is required.

Simple Example

Smart Healthcare System

Opportunities:

- real-time patient monitoring
- low latency alerts

Challenges:

- securing fog devices
- managing distributed nodes
- ensuring reliability

Both must be addressed together.

Fog Computing Communication Technologies

Introduction

Fog computing connects:

- IoT devices
- fog nodes
- cloud data centers

To work efficiently, fog computing depends on **communication technologies** that enable:

- data transfer
- control signals
- real-time interaction

Communication is the backbone of fog computing.

Why Communication Technologies are Important?

Fog environments are:

- distributed
- heterogeneous
- latency-sensitive

Communication technologies help to:

- move data between devices and fog nodes
- reduce latency
- support real-time applications
- ensure reliable connectivity

Communication Layers in Fog Computing

Fog computing communication happens across **three layers**:

- Device-to-Fog (Edge Communication)
- Fog-to-Fog (Horizontal Communication)
- Fog-to-Cloud (Vertical Communication)

Each layer uses different communication technologies.

Device-to-Fog Communication

This communication occurs between:

- sensors
- actuators
- IoT devices
- nearby fog nodes or gateways

Characteristics:

- short range
- low power
- frequent data transmission

Technologies for Device-to-Fog Communication

Common technologies include:

- Bluetooth / Bluetooth Low Energy (BLE)
- ZigBee
- Z-Wave
- IEEE 802.15.4
- Wi-Fi

These technologies support energy-efficient IoT communication.

Fog-to-Fog Communication

- Fog-to-fog communication connects:
- fog nodes at the same layer
- neighboring gateways
- edge servers

Purpose:

- data sharing
- load balancing
- coordination between fog nodes

Enables distributed fog processing.

Technologies for Fog-to-Fog Communication

Common technologies include:

- Ethernet
- Wi-Fi
- TCP/IP networks
- Software Defined Networking (SDN)

Supports higher bandwidth and faster data exchange.

Fog-to-Cloud Communication

Fog-to-cloud communication connects:

- fog nodes
- central cloud data centers

Purpose:

- long-term storage
- large-scale analytics
- global system management

Cloud handles tasks not suitable for fog.

Technologies for Fog-to-Cloud Communication

Common technologies include:

- Internet (IP-based communication)
- TCP / UDP
- HTTP / HTTPS
- VPN connections

Secure and reliable communication is required.

Communication Protocols in Fog Computing

- Fog systems use **application-layer protocols** for data exchange.

Common protocols:

- HTTP / HTTPS
- MQTT
- CoAP
- AMQP
- XMPP

Choice depends on latency, bandwidth, and device capability.

MQTT Protocol

- **MQTT (Message Queue Telemetry Transport):**
- lightweight
- publish/subscribe model
- low bandwidth usage

Used in:

- IoT
- fog-based real-time monitoring
- Suitable for constrained devices.

CoAP Protocol

CoAP (Constrained Application Protocol):

- designed for constrained devices
- works over UDP
- request/response model

Used in:

- smart sensors
- low-power networks

Efficient for fog-edge communication.

HTTP/HTTPS in Fog Computing

HTTP/HTTPS is used for:

- fog-to-cloud communication
- REST APIs
- service management

Advantages:

- widely supported
- easy integration
- Limitation:
- higher overhead compared to MQTT or CoAP

Networking Technologies Supporting Fog

Fog communication is supported by:

- Ethernet
- Wi-Fi
- Cellular networks (4G / 5G)
- Local Area Networks (LAN)
- Wide Area Networks (WAN)

5G plays a major role in fog computing.

Role of 5G in Fog Communication

5G enables:

- ultra-low latency
- high bandwidth
- massive device connectivity

Applications:

- autonomous vehicles
- smart cities
- AR/VR
- industrial IoT

5G strengthens fog deployment.

Security in Fog Communication

- Communication security is essential.
- Security measures include:
 - encryption
 - authentication
 - authorization
 - secure protocols (HTTPS, TLS)

Unsecured communication can lead to attacks.

Challenges in Fog Communication

Communication challenges include:

- network heterogeneity
- latency variations
- bandwidth limitations
- mobility of devices
- security threats

Efficient protocol selection is required.

Simple Example

Smart City Scenario

- Sensors → ZigBee / BLE
- Fog gateway → Wi-Fi / Ethernet
- Cloud → Internet / HTTPS

Different layers use different communication technologies.